

Geometry
Unit 13
Lesson 3 Practice

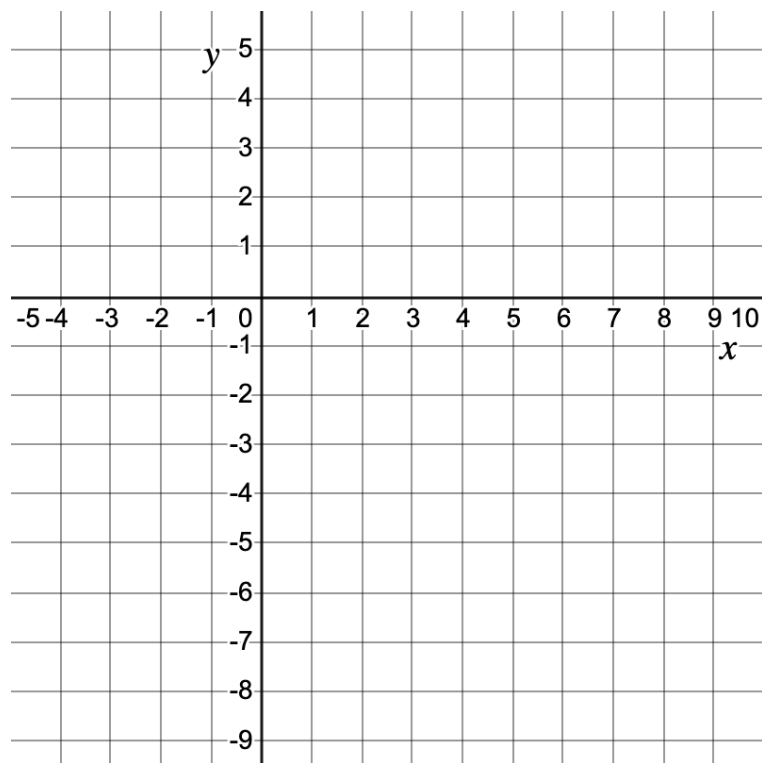
Name _____
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The equation of circle C is $(x - 6)^2 + (y + 3)^2 = 9$.

1. Complete the table by substituting the given coordinates for x and y in the equation of circle C . Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$R(6,0)$		_____ = 9
$G(3,-3)$		_____ = 9
$T(6,-6)$		_____ = 9

2. Graph circle C , $(x - 6)^2 + (y + 3)^2 = 9$.



3. Plot $R(6,0)$, $G(3,-3)$, $T(6,-6)$ on the circle.
4. Determine the length of the diameter on circle C .

Circle D has a radius of 2 with points $L(-3,7)$, $F(-1,5)$, and $T(1,7)$ on the circle.

5. Determine which two points are the endpoints of a diameter of circle D .
6. Find the center of circle D .
7. Write the equation for circle D .
8. The equation of circle W is $(x + 7)^2 + y^2 = 4$. Determine whether the following points lie inside, outside, or on circle W .

Point	Work	Location
$C(-7,2)$		☐ Inside circle W ☐ Outside circle W ☐ On circle W
$N(-6,1)$		☐ Inside circle W ☐ Outside circle W ☐ On circle W
$T(-7,5)$		☐ Inside circle W ☐ Outside circle W ☐ On circle W

9. The equation of circle R is $(x - 4)^2 + (y + 2)^2 = 13$.

Select all points that lie on circle R .

- ☐ $(6,1)$
 - ☐ $(4,-2)$
 - ☐ $(1,0)$
 - ☐ $(0,0)$
 - ☐ $(-4,-2)$
10. Circle M and circle T intersect at point H .

Equation of circle M is $x^2 + (y - 2)^2 = 10$

Equation of circle T is $(x + 4)^2 + (y - 2)^2 = 2$

Which coordinate pair is the location of point H .

- ☐ $(-5,2)$
- ☐ $(0,0)$
- ☐ $(-3,0)$
- ☐ $(-3,1)$